

T-97701 C4MBI67 CRITICAL CORE

Actual publication recovery for the missing T-97701 materials.
Status: exact identities proved; C4MBI67 sign open; RH unproved.
Base: PR 590 at 4f1283c67f0d4a4b504badd4c4113ba227521162.
Old T-97700 ZIP BF79073D...3FF640 is not reused.

1. ANNULAR SCALAR

Let $R_X = 5 c_X^2 + 3 c_X^3$ and $A_X = R_X - R_{Xover4}$.
Let $H_Y n = \min \log_4, \log Y$ overn positive part.
 $a_star n = 6 one_n1 - 6 \mu n + 9 one_2divn \mu ndiv2 - 3 one_4divn \mu ndiv4$.
 $kappa_X n = -6 H_X n + 9 \text{oversqrt}2 H_{Xover2} n - 3 \text{over}2 H_{Xover4} n$.
Exact identity:
 $A_X = 6 H_X 1 + C_X$,
 $C_X = \sum \mu n kappa_X n \text{oversqrt} n$.

2. EXACT FOUR BAND BOUNDARY

Put $B_half t = \sum n \text{lesst} B \mu n \text{oversqrt} n$.
 C_X equals:
 $-3 \text{over}2 \int_{Xover16}^{Xover8} B_half t dt \text{over}t$,
plus $9 \text{sqrt}2 \text{minus}3 \text{over}2 \int_{Xover8}^{Xover4} B_half t dt \text{over}t$,
plus $9 \text{sqrt}2 \text{minus}12 \text{over}2 \int_{Xover4}^{Xover2} B_half t dt \text{over}t$,
minus $6 \int_{Xover2}^X B_half t dt \text{over}t$.
This is exact for every real X greater than zero.
Define C4MBI67 X by C_X greater than or equal to $-6 H_X 1$.
Then C4MBI67 X if and only if A_X is nonnegative.

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3. ROOT BLPTE67 EQUIVALENCE

Retain the exact T-97700 Bellman identity:

$$U_{\text{full}} X = U_Z X - I_Z X - T_Z X.$$

At the root $U_{\text{full}} X = A_X \text{ oversqrt } X$.

BLPTE67 X is $T_Z X$ less than or equal to $U_Z X - I_Z X$.

Hence pointwise at the root:

BLPTE67 X iff $U_{\text{full}} X$ nonnegative iff A_X nonnegative iff C4MBI67 X .

This does not silently assert the stronger state-wise quantifiers.

4. EXACT PRIME MOBIUS OWNERSHIP

For every squarefree n greater than one:

$\mu n \log n = \text{minus sum over primes } p \text{ dividing } n \text{ of } \mu \text{ ndivp } \log p$.

Therefore:

$\text{sum } n \text{ greater1 } \mu n \text{ kappa_X } n \text{ oversqrt } n \text{ equals}$

$\text{minus sum } p \text{ lessthan } X \log p \text{ oversqrt } p \text{ times}$

$\text{sum } m \text{ lessthan } X \text{ overp and } p \text{ not dividing } m \text{ of}$

$\mu m \text{ kappa_X } p m \text{ over sqrt } m \log p m$.

Each squarefree source atom is spent exactly once with owner weights $\log p$ over $\log n$.

5. UNSIGNED TYPE II FIREWALL

Two equal magnitude signed atoms with distinct kernel values can preserve support, l_1 , l_2 , unsigned owner mass, and source-blind large-sieve data while reversing the one-sided functional. Thus unsigned data cannot decide C4MBI67.

This refutes a proof mechanism, not the actual Mobius inequality.

6. REPLAY AND THEOREM LEDGER

Checker verdict PASS_T97701_BLPTE_C4MBI67_CRITICAL_CORE_ALGEBRA.

Proof object 8c65f8e1a340510c153688196de0782938eb377277cc425808b53e5d931b1f6d.

Four-band identity: PROVED EXACT.

Prime-Mobius ownership: PROVED EXACT.

BLPTE67 C4MBI67 annular equivalence: PROVED EXACT AT ROOT.

Source-blind unsigned Type-II implication: REFUTED AS A MECHANISM.

C4MBI67 sign: OPEN AND RH-BEARING.

Riemann Hypothesis: UNPROVED.